Perception Algorithm

- dataset D_n with $x \in \mathbb{R}^d$, $y \in \{-1, 1\}$
- $x^{(i)} \longrightarrow x^{(i)}_{new} = [x_1, \ldots, x_d, 1]$

  then remove offset θ_0 from classifier
- $\theta = 0$
- repeat if $y^{(i)}(\theta^T x^{(i)}) \leq 0$

  then $\theta \mathrel{+}= y^{(i)} x^{(i)}$

Note.

$$h(x;\theta) = \begin{cases} +1 & \text{if } \theta^T x \geq 0 \\ -1 & \text{otherwise} \end{cases}$$

D_n is linearly separable

if $\exists\, \theta, \theta_0$ s.t $\forall i$ $y^{(i)}(\theta^T x^{(i)} + \theta_0) > 0$

i.e $\varepsilon_n(h) = 0$

(training error)

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hyperplane θ, θ_0

(i) distance $\dfrac{\theta^T x + \theta_0}{\|\theta\|}$ $\Longrightarrow$ margin $y \cdot \dfrac{\theta^T x + \theta_0}{\|\theta\|}$

(ii) margin of dataset D_n $\min_i \left(y^{(i)} \dfrac{\theta^T x^{(i)} + \theta_0}{\|\theta\|} \right)$

i.e margin > 0

if classified correctly

Perceptron Convergence Thm. $\Longrightarrow$ at most

(i) $\exists \theta^*$ s.t $y^{(i)} \dfrac{\theta^{*T} x^{(i)}}{\|\theta^*\|} \geq \gamma$

(ii) $\|x^{(i)}\| \leq R$

$$\left(\frac{R}{\gamma}\right)^2 = \left(\frac{\text{데이터의 반경}}{\text{정답의 margin}}\right)^2 \text{ mistakes}$$

$$\cos(\theta^{(k)}, \theta^*) = \left(\frac{\theta^{(k)} \cdot \theta^*}{\|\theta^*\|}\right)\left(\frac{1}{\|\theta^{(k)}\|}\right) \leq 1$$

(i) $$\frac{\theta^{(k)} \cdot \theta^*}{\|\theta^*\|} = \frac{(\theta^{(k-1)} + y^{(i)} x^{(i)}) \cdot \theta^*}{\|\theta^*\|} = \underbrace{\frac{\theta^{(k-1)} \cdot \theta^*}{\|\theta^*\|}}_{\geq (k-1)\gamma} + \underbrace{y^{(i)} \frac{x^{(i)} \cdot \theta^*}{\|\theta^*\|}}_{\geq \gamma} \geq k\gamma$$

(ii) $$\|\theta^{(k)}\|^2 = \|\theta^{(k-1)} + y^{(i)} x^{(i)}\|^2$$

$$= \|\theta^{(k-1)}\|^2 + 2y^{(i)} \theta^{(k-1)T} x^{(i)} + \|x^{(i)}\|^2$$

$$\leq \underbrace{\|\theta^{(k-1)}\|^2}_{\leq (k-1)R^2} + \underbrace{\|x^{(i)}\|^2}_{\leq R^2} \leq kR^2$$

$$k\gamma \cdot \frac{1}{\sqrt{kR^2}} \leq \left(\frac{\theta^{(k)} \cdot \theta^*}{\|\theta^*\|}\right) \cdot \left(\frac{1}{\|\theta^{(k)}\|}\right) \leq 1 \quad \text{implies} \quad k \leq \left(\frac{R}{\gamma}\right)^2$$

Linear Classifier

(Linear Regression) $h_\theta(x) = \theta^T x$

(feature map) $h_\theta(x) = \theta^T \phi(x)$

Note.

실제 데이터의 입출력은 근본적으로 비선형적이다.

Regression

- $x^{(i)} \longrightarrow y^{(i)} \in \mathbb{R}$
- $J^{(i)}(\theta) = \frac{1}{2}\left(h_\theta(x^{(i)}) - y^{(i)}\right)^2$
- $J(\theta) = \frac{1}{n}\sum_{i=1}^{n} J^{(i)}(\theta)$

i.e Mean Squared Error

Binary Classification

- logit : $\bar{h}_\theta : \mathbb{R}^d \longrightarrow \mathbb{R}$
- probability : $h_\theta(x) = \sigma(\bar{h}_\theta(x))$ i.e sigmoid

$$= \frac{1}{1+e^{-\bar{h}_\theta(x)}}$$

- binary cross-entropy loss

$$J(\theta) = -\frac{1}{n}\sum_{i=1}^{n}\left[y^{(i)} \log h_\theta(x^{(i)}) + (1-y^{(i)}) \log(1-h_\theta(x^{(i)}))\right]$$

Multi-class Classification

- logit $\bar{h}_\theta(x) = \left[\bar{h}_{\theta,1}(x) \ldots \bar{h}_{\theta,k}(x)\right] \in \mathbb{R}^k$
- softmax $\hat{y}_c^{(i)} = P(y=c \mid x^{(i)} : \theta) = \dfrac{e^{\bar{h}_{\theta,c}(x^{(i)})}}{\sum_s e^{\bar{h}_{\theta,s}(x^{(i)})}}$
- Cross-Entropy Loss $-\frac{1}{n}\sum_{i=1}^{n}\sum_{c=1}^{k} y_c^{(i)} \log \hat{y}_c^{(i)}$

Neural Networks

(i) Single Neuron : $\bar{h}_\theta(x) = \mathrm{ReLU}(w^T x + b)$

(ii) hidden unit : $a_1 = \mathrm{ReLU}(\theta_1 x_1 + \theta_2 x_2 + \theta_3)$

$$\bar{h}_\theta(x) = \theta_9 a_1 + \theta_{10} a_2 + \theta_{11} a_3 + \theta_{12}$$

(iii) Multi Layer

Fully Connected NN

$$a^{(1)} = \mathrm{ReLU}(W^{(1)} x + b^{(1)})$$

$$\cdots$$

$$a^{(L-1)} = \mathrm{ReLU}(W^{(L-1)} a^{(L-2)} + b^{(L-1)})$$

$$\bar{h}_\theta(x) = W^{(L)} a^{(L-1)} + b^{(L)}$$

Activation Funcs

sigmoid $\frac{1}{1+e^{-z}}$

tanh $\frac{e^z - e^{-z}}{e^z + e^{-z}}$

leaky ReLU $\max\{z, \gamma z\}$, $\gamma \in (0,1)$

GELU $\frac{z}{2}\left(1 + \mathrm{erf}\left(\frac{z}{\sqrt{2}}\right)\right)$

softplus $\frac{1}{\beta} \log(1 + \exp(\beta z))$

Deep Learning Automatically Learns Feature Map.

$\bar{h}_\theta(x) = W \phi_\beta(x) + b$

① linear param $W \cdot b$

② non-linear feature map ϕ_β

acc-1 신경망은

중간에 리턴함.

Residual Network

Note.

$MM(x) = Wx + b$

(before) $\sigma(MM(z))$

(after) $Res(z) = z + \sigma(MM(\sigma(MM(z))))$

$\Rightarrow ResNet\text{-}S(x) = MM(Res(Res(\cdots Res(x))))$

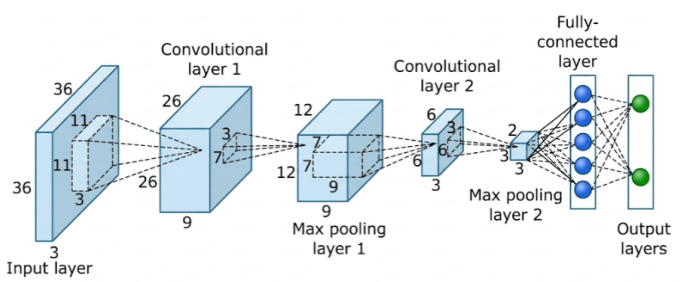
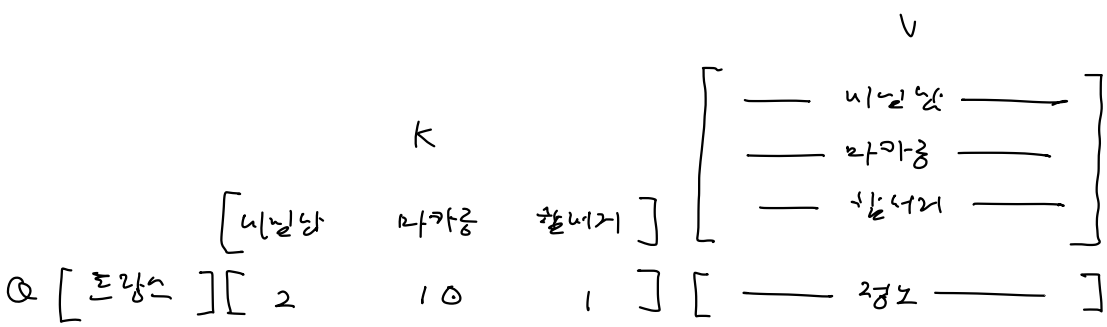


Fig. A hierarchical 2D convolutional neural network pipeline

$$\text{Attention} = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$



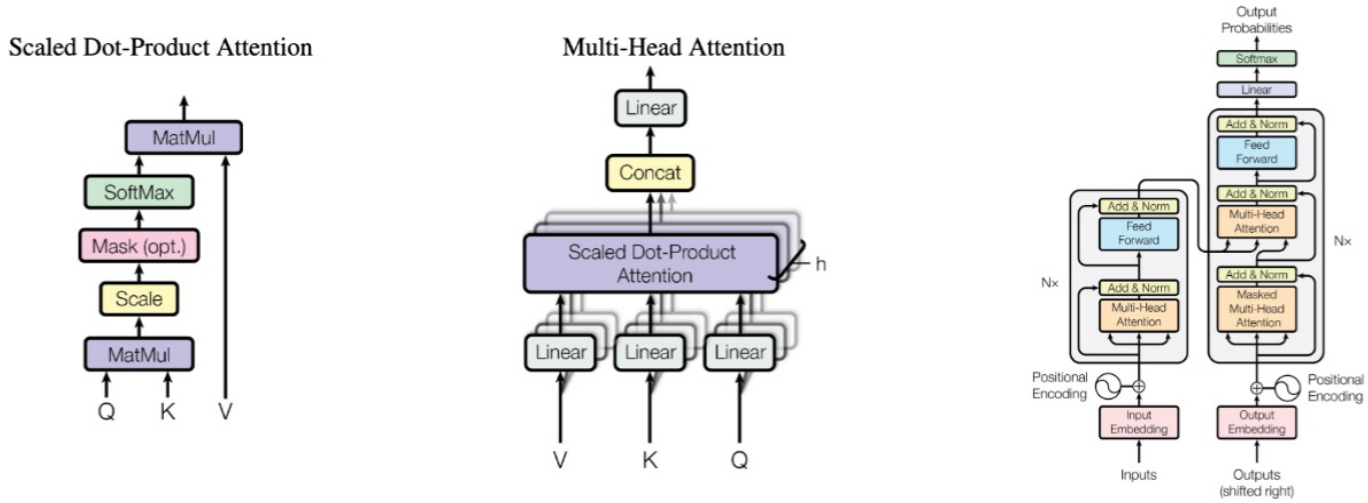


Fig. Self-attention and transformer architecture

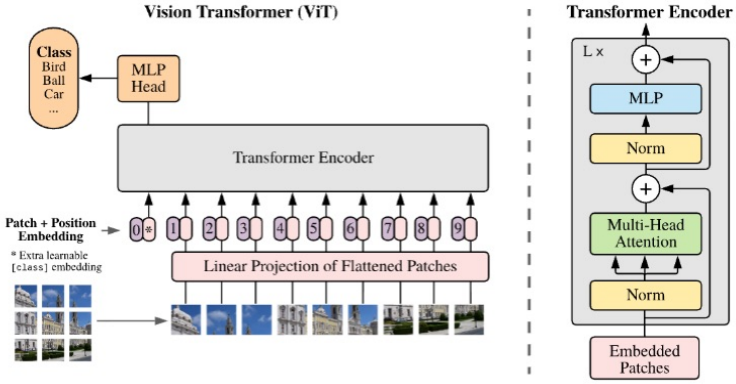
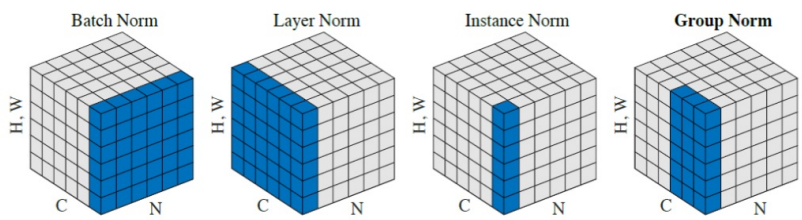


Fig. Vision transformer



Layer normalization

- Normalize hidden features for each sample
- Mean and variance computed within the sample
- Independent of mini-batch statistics

$$\mathrm{LN}(x) = \gamma \odot \frac{x - \mu}{\sqrt{\sigma^2 + \epsilon}} + \beta$$